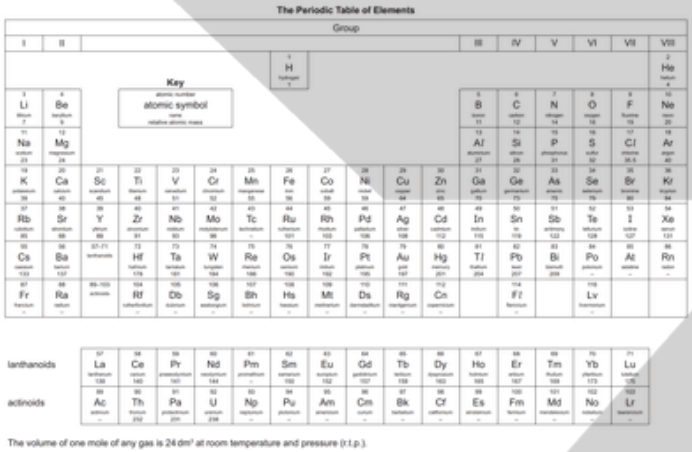
1. The Periodic Table

1.1. Arrangement of Elements

- The Periodic table is a method of classifying elements.
- Elements are arranged in order of increasing atomic number (each proceeding element has one more proton)
- Made up of rows called periods and columns called **groups**, the position of an element helps determine its electronic configuration.
- **Period number**: number of electron shells
- **Group number**: number of outer shell electrons

Elements in the same group have similar chemical properties.



Periodic Trends

1. The table moves from metals on the left to non-metals on the right.
2. Down a group of metals, elements become more reactive.
3. With non-metals going down a group, reactivity decreases.

1.2. Group I Properties

- Group I metals: Lithium, sodium and potassium

Chemical Properties	Physical Properties
These metals must be stored in oil to prevent them reacting with oxygen and water in the air.	Good conductors of heat and electricity
React violently with chlorine	Soft and easy to cut
Burst into flames when heated with oxygen[red flame for lithium; yellow flame for sodium; lilac flame for potassium]	Shiny when freshly cut
Produce soluble white compounds.	Melting points decrease down the group.
React with water to form alkaline metal and hydrogen gas	Increase density down the group

- Predicting the properties of other Group I alkali metals: Rubidium, Caesium and Francium [reactivity increases down the group]

Element	Reaction with Water
Lithium	Floats and gives off hydrogen gas (effervescence)
Sodium	Vigorous Reaction and moves very quickly
Potassium	Explosive Reaction, lilac flame

1.3. Group VII Properties

Properties	Patterns
States and Colours, at RTP: Fluorine- Yellow-green gas Chlorine- Pale Yellow-Green gas Bromine- Red-Brown liquid Iodine- Grey-Black solid	Down the group, size, mass and density increase
Poisonous	Down the group, the colour darkens
Diatomic; form halide ions in a displacement reaction	Reactivity decreases down the group, because it has to gain an electron, so the closer the electron is to the positive nucleus, the more easily it will be gained, so atoms with fewer shells will react more easily.
Do not conduct electricity	melting point increases down the group

1.4. Transition Elements

Physical Properties

- High melting & boiling points.
- Malleable and ductile
- Good conductors of heat & electricity
- High density

Chemical Properties

- Act as catalysts
- Form coloured compounds
- Variable Oxidation Numbers (Iron (II) or Iron (III))

1.5. Noble Gases

Properties	Uses
Density increases down the group	Helium-filling balloons
Monoatomic and colourless	Argon – Lamps
M.P. and B.P. increases down the group	Neon – advertising signs
Don't conduct electricity	
Inert & stable due to full outer shell electrons	

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